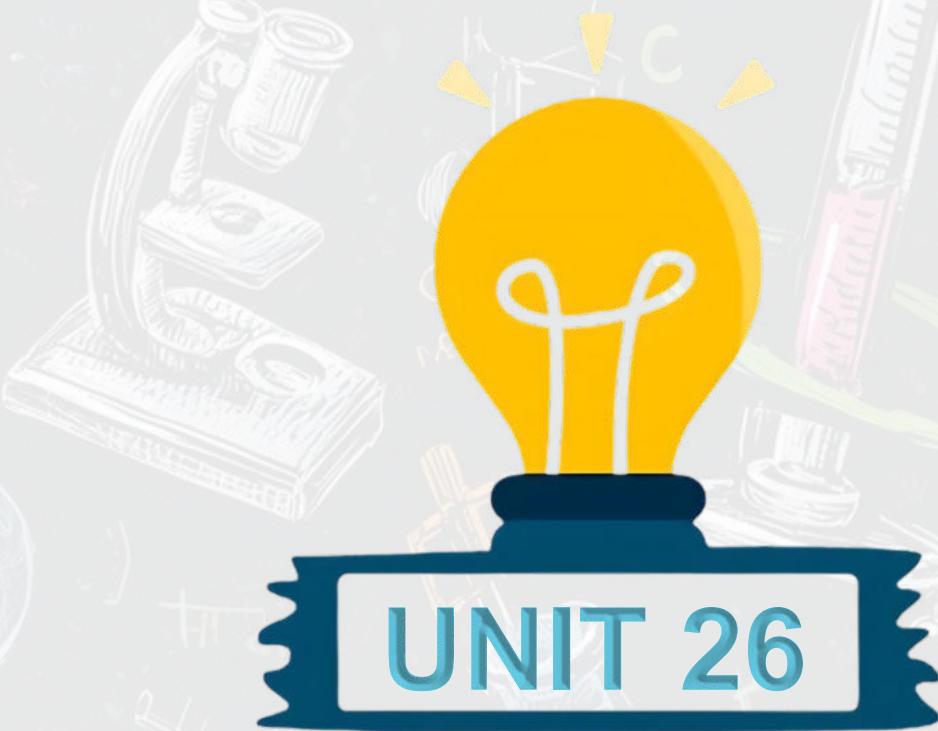
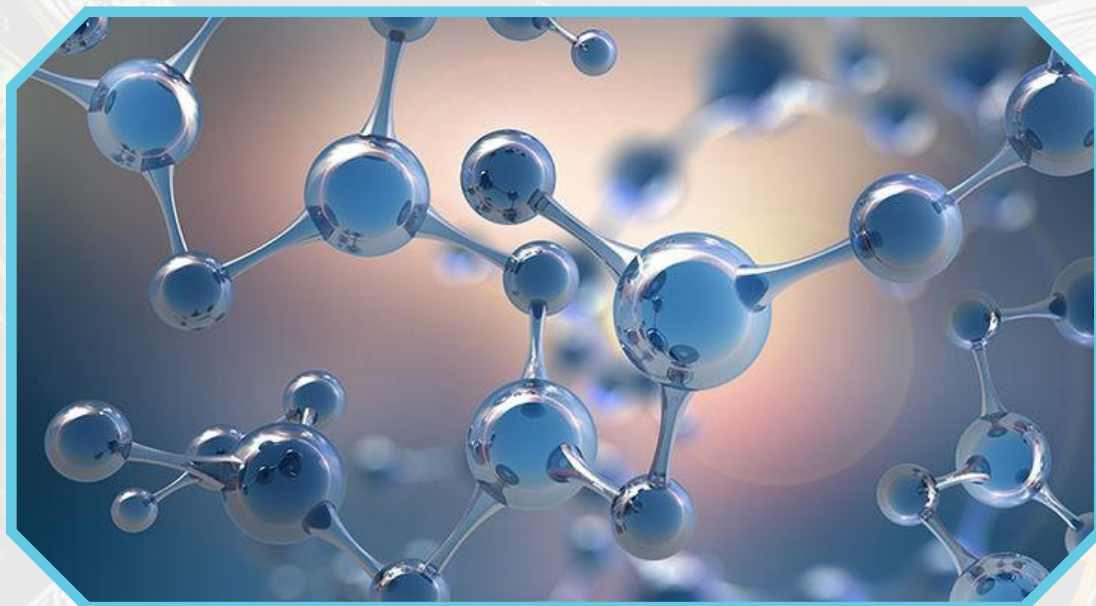




UNIT 26

ATOMIC SPECTRA



Atomic Spectra:

Atomic spectra deal with the spectrum of frequencies of electromagnetic radiation emitted or absorbed by an electron during transitions between different levels of energy within an atom.

Optical Spectra:

An optical spectrum is obtained when light passes through a prism or a diffraction grating. This spectrum can be obtained from the intensity and wavelength of the radiation.

(i) Continuous spectra:

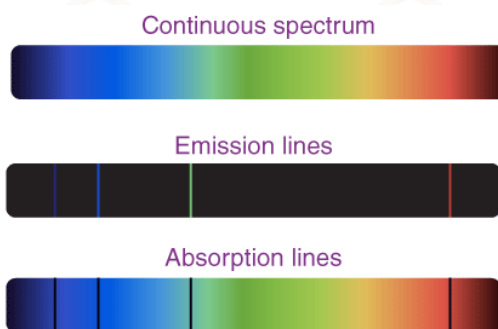
These are emitted by solids, liquids, and dense opaque gases at high temperatures. The spectrum of the sun, or of a black body, is a continuous spectrum. Gases at low pressures emit line spectra.

(ii) Band spectra:

These are associated with similar changes in the molecules under high pressure. Emission lines.

(iii) Line spectra:

These have their origin in the energy changes which take place in the atoms of a gas.



Spectral lines:

A line of particular frequency or wavelength emitted or absorbed by atoms is called a spectral line.

The four visible lines occur at the wavelengths 656 nm, 486 nm, 434 nm, and 410 nm. The complete set of lines is called the Balmer series. The wavelengths of these lines can be between 1860 and 1885, described by the following equation:

$$\frac{1}{\lambda} = R_H \left(\frac{1}{2^2} - \frac{1}{n^2} \right); n = 3, 4, 5, \dots, 26.1$$

1. Balmer Series $\frac{1}{\lambda} = R_H \left(\frac{1}{2^2} - \frac{1}{n^2} \right)$

2. Paschen Series $\frac{1}{\lambda} = R_H \left(\frac{1}{3^2} - \frac{1}{n^2} \right)$

3. Lyman Series $\frac{1}{\lambda} = R_H \left(\frac{1}{1^2} - \frac{1}{n^2} \right)$

4. Brackett Series $\frac{1}{\lambda} = R_H \left(\frac{1}{4^2} - \frac{1}{n^2} \right)$

5. PFund Series $\frac{1}{\lambda} = R_H \left(\frac{1}{5^2} - \frac{1}{n^2} \right)$

Identification of Elements by Uniqueness of Spectra:

Each element has its own unique line spectrum which is referred to as the "fingerprint" of a particular element.

Bohr's Model and its Postulates:

In 1913, Niels Bohr introduced the atomic model in order to give quantitative determination of frequency emitted during de-excitation of an electron in the Hydrogen atom.

The discovery of the electron in the late 19th century marked the beginning of a new era in scientific research, helping physicists understand the structure and nature of atoms.

Niels Bohr had a remarkable idea that explained the atom of hydrogen - consist of a proton and an electron revolving around the nucleus.

The following are the postulates of Bohr:

Postulates I:

The electron in a hydrogen atom orbits the nucleus,

$$F_c = \frac{mv^2}{r} = F = k \times \frac{q_1 q_2}{r^2}$$

Postulates II:

The magnitude of the angular momentum L of the electron in its orbit is equal to the integral multiple of $\frac{h}{2\pi}$ i.e.,

$$L = n \frac{h}{2\pi}, n = 1, 2, 3, 4, \dots$$

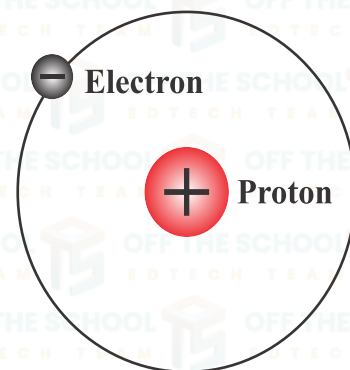
where h is called Plank's constant and n is positive integer (quantum number).

Postulates III:

Only certain orbits are stable in which electrons are revolving and these orbits are called stationary states. The atom emits radiation (photon) when the electron makes a transition from a higher energy state (E_n) to the lower energy state (E_p).

$$h\nu = E_n - E_p$$

where ν is the frequency of emitted photon.



hydrogen atom

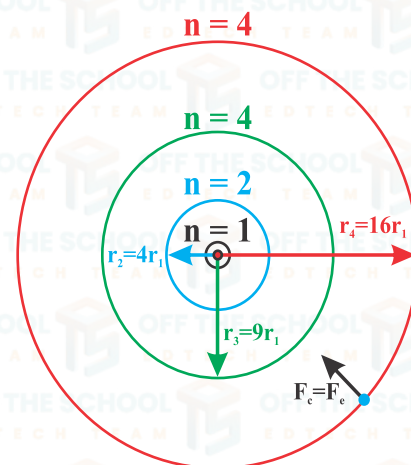
26.2.2 Bohr's radius:

The electron revolving around the nucleus is in uniform circular motion and thus experiences a centripetal force. Here centripetal force is provided by electrostatic force between electron and proton.

$$F_c = \frac{mv^2}{r} \quad (i)$$

where V is the speed of electron. The magnitude of electrostatic force ($F_e = \frac{kq_1q_2}{r^2}$) between the electron ($q_1 = -e$) and the proton ($q_2 = e$) separated by the orbital radius r is given as.

$$F_e = \frac{ke^2}{r^2} \quad (ii)$$



quantized radii for hydrogen atom

where $k = 9.0 \times 10^9 \text{ Nm}^2/\text{C}^2$ is called Coulomb's constant.

The electron can only move in a particular orbit if the above two forces are balanced each other.

Comparing equation (i) and (ii)

$$\frac{ke^2}{r^2} = \frac{mV^2}{r} \quad (\text{iii})$$

From Bohr's postulate (2),

$$L = mVr = n \frac{h}{2\pi}$$

Substituting the value of $V = \frac{nh}{2\pi mr}$

$$\frac{ke^2}{r^2} = \frac{m}{r} \left(\frac{nh}{2\pi mr} \right)^2$$

Where,

$$k = \frac{1}{4\pi\epsilon_0}$$

$$h = 6.63 \times 10^{-34} \text{ Js}$$

$$m = 9.1 \times 10^{-31} \text{ kg}$$

$$\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2/\text{Nm}^2$$

$$r = \frac{n^2 \epsilon_0 h^2}{\pi m e^2} \quad (\text{IV})$$

$$r = r_0 n^2 \quad (\text{v})$$

where $r_0 = 5.29 \times 10^{-11} \text{ m} = 0.53 \text{ \AA}$ is called Bohr's first orbit.

As the values of r is depending only on the principle quantum number n . Therefore, the radii have the quantized values

$$r_n = r_1 n^2$$

26.2.3: Orbital Energy of hydrogen atom:

The allowed energy levels and quantitative values of the emission wavelengths of the hydrogen atom can be calculated from the postulate (3) suggests the qualitative existence of a characteristic discrete emission spectrum and corresponding absorption spectrum for hydrogen. The electron has kinetic energy ($K = \frac{1}{2} mV^2$) and electric potential energy ($U = -\frac{kq_1q_2}{r}$). The total energy will be

$$E = K + U$$

$$E = \frac{1}{2} mV^2 - \frac{ke^2}{r}$$

Putting the value of $mV^2 = \frac{ke^2}{r}$ from equation (III)

$$E = \frac{ke^2}{2r} - \frac{ke^2}{r}$$

$$E = -\frac{ke^2}{2}$$

$n=\infty$	-----	$E=0$
$n=4$	=====	$E=-0.85\text{eV}$
$n=3$	-----	$E=-1.51\text{eV}$
$n=2$	-----	$E=-3.40\text{eV}$
$n=1$	-----	$E=-13.6\text{eV}$

various excitation levels

Substitute the value of r from equation (IV)

$$E = -\frac{me^4}{8\varepsilon_0^2 h^2} \left(\frac{1}{n^2} \right)$$

The numerical value of this energy will be

$$E_n = \frac{-13.6}{n^2} \text{ eV}$$

The energy level diagram shows the various -excitation levels.

X-Rays:

These are powerful electromagnetic rays of higher energy and can pass through most objects, including the body.

Most of them have a wavelength ranging from 0.01 to 10 nanometers, corresponding to 3×10^{19} Hz to 3×10^{16} Hz.

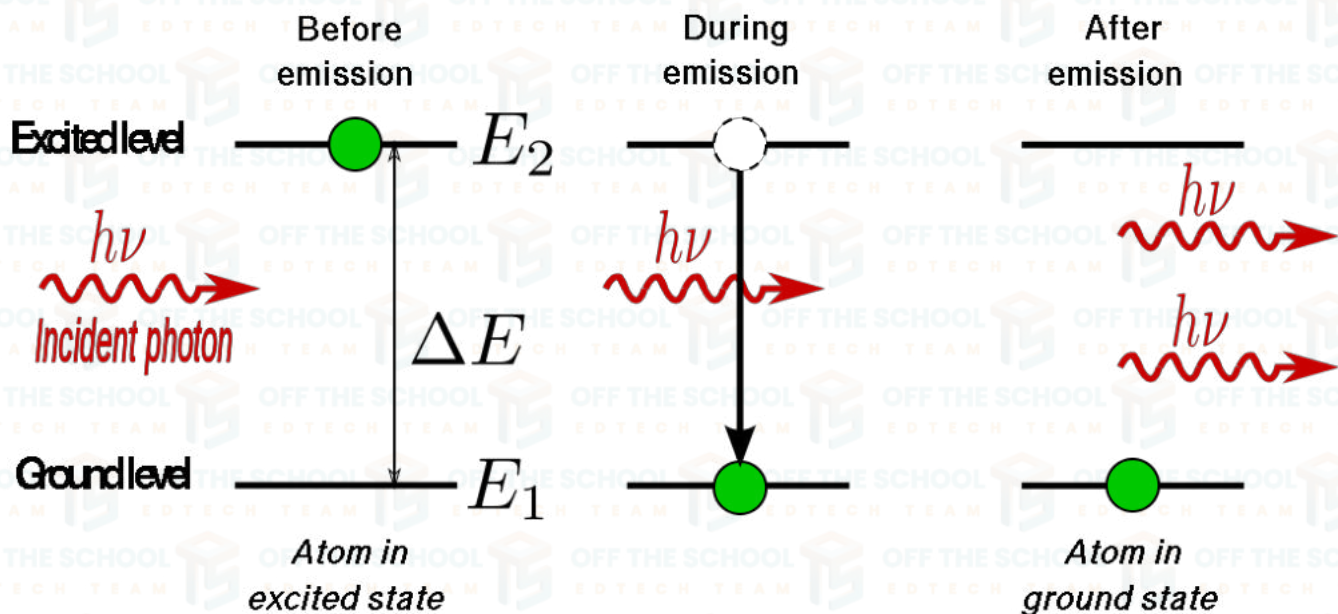
X-rays are used to generate images of tissues and structures inside the body.

LASER: Light Amplification by Stimulated Emission of Radiation:

Lasers are designed to produce and amplify this stimulated form of light into intense and focused beams. Compared to conventional sources of ordinary light, the light from a laser is quite intense, monochromatic, and emitted in a unidirectional beam limited by diffraction.

Stimulated or Induced Absorption:

When an atom induced by the photon (energy packet) and do transition to one of its allowed states (excited state) called Stimulated Or induced absorption.



$$E_2 - E_1 = \Delta E = h\nu$$

Spontaneous Emission:

The process of photon emission by an excited atom with no external influences is called spontaneous emission.

Stimulated Emission:

Radiation, which occurs as a result of external exposure, is called induced or stimulated. In the stimulated emission two photons are involved: the primary photon, causing the emission of radiation by an excited atom and the secondary photon emitted by the atom.

Population Inversion:

Population inversion is a key to produce laser light.

The process of achieving a greater population in a higher energy state compared to a lower energy state.

Population inversion cannot be achieved in a two energy level system, at normal conditions; the number of electrons in the lower energy state (E_1) is always greater as compared to the number of electrons in the higher energy state (E_2).

Metastable States:

Consider a system consisting of three energy levels E_1 , E_2 , E_3 . We assume that the energy level of E_1 is less than E_2 and E_3 , the energy level of E_2 is greater than E_1 and less than E_3 , and the energy level of E_3 is greater than E_1 and E_2 . The energy level E_2 is sometimes referred to as Metastable state having a life time of 10 seconds.

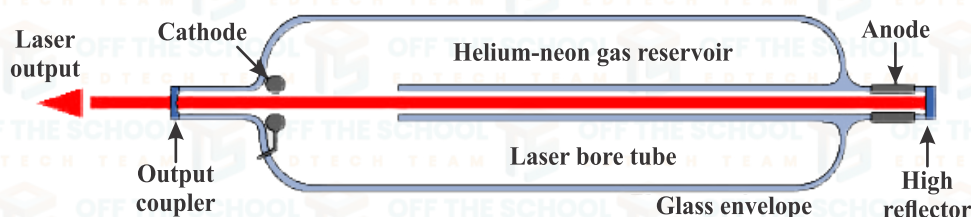
He-Ne Gas Laser:

The helium-neon laser was the first continuous wave (CW) laser ever constructed.

It was built in 1961 by Ali Javan, Bennett, and Herriott at Bell Telephone Laboratories.

He-Ne lasers are commonly used in school laboratories and in older barcode readers.

The schematic diagram of He-Ne laser is shown below.



Helium-neon laser construction:

The helium-neon laser consists of three essential components:

Pump Source (high voltage power supply):

The pump energy of the laser is provided by an electrical discharge of several hundred Volts between an anode and cathode at each end of the glass tube.

A current of 5 to 100mA is typical for laser operation.

Gain Medium (laser glass tube or discharge glass tube):

Figure shows a gas discharge tube containing a low-pressure mixture of helium-neon in a ratio between 5:1 and 20:1 bound in a glass tube. The partial pressure of helium is 1 mbar whereas that of neon is 0.1 mbar.

Resonating Cavity:

The glass tube (containing a mixture of helium and neon gas) is placed between two parallel mirrors. These two mirrors are silvered or optically coated. Each mirror is silvered differently. The left side mirror is partially silvered and is known as the output coupler whereas the right-side mirror is fully silvered and is known as the high reflector or fully reflecting mirror. The fully silvered mirror will completely reflect whereas the partially silvered mirror will reflect most part of the light but allows some part of the light to produce the laser beam.

Laser Operation:

This electrical excitation raises the helium atom into a meta-stable state with energy 20.61 eV above the ground state. Neon has a meta-stable state with energy 20.66 eV above its ground state. Collision of the excited helium atoms with the ground-state neon atoms results in transfer of energy to the neon atoms, exciting neon electrons. The difference between the energy states of the two atoms is in the order of 0.05 eV, which is supplied by kinetic energy.

The number of neon atoms in the excited states builds up as further collisions between helium and neon atoms occur, causing a population inversion. Spontaneous and stimulated emission results in emission of 632.82 nm wavelength light.

Properties of Laser:

The properties of laser light are very strange as compared to conventional light. The factors that make laser light prominent are: Monochromatic, Highly Intense or Brightness, Coherence and Directionality.

1. Monochromatic:

A single color or wavelength light is called monochromatic. The light emitted from ordinary light sources has different wavelengths, or colors. However, laser light has a single wavelength or color.

2. Highly Intense or Brightness:

The characteristics of coherence and directionality of the laser makes laser light highly intense as compared to conventional light. It is the power emitted per unit surface area per unit solid angle. A one milliwatt He-Ne laser is more intense than the sun intensity.

3. Coherence:

Two or more waves of same frequency are said to be coherent in nature if they have constant phase difference. A predictable correlation of the amplitude and phase at any one point with another point is called coherence. In laser light the property of coherence occurs between any two or more light Coherence waves. Whereas in conventional light, the property, coherence exhibits between a source and its virtual source.

4. Directionality or Divergence:

The light ray coming from laser light travels in a single direction. However, an ordinary light source travels in all directions. For example, on travelling a distance of 1 Km the torch light spreads 1 km distance. But the laser light spreads only few centimeters distance even it travels lacks kilometre distance.

Uses of Lasers:

Lasers are widely used in many fields, some of them are:

1. Tools:

- Cutting tools are typically precise, and also simple to automate. They do not require sharpening, unlike that of knives.
- Robot-guided lasers are widely used for the cutting of pieces of cloth.
- Laser tools are used in corneal surgery, restoring a detached retina of eye, and removing kidney stones.

2. Communication:

- Lasers are used in Barcode scanners to convert a printed barcode into a number.
- A semiconductor laser beam helps to convert the printed pattern of data into numbers in a CD/DVD.
- In Photonics, lasers are used in fiber optic cables.

3. Defence:

- In defence field, the military employs laser-guided guns and missiles.
- Laser range-finders use high-resolution scanning to find the distance and speed from an object that is located beyond the point-blank range.

4. Medicine:

- In medicine, the laser beam is used as part of phototherapy for many procedures.
- The development of laser technology in recent decades has enabled the creation of a new field of medicine laser surgery.
- Lasers have uses in dermatology, ophthalmology, urology, rheumatology and dentistry.

5. Holography:

- Holography is a true three-dimensional image recording on film by lasers. Holograms are used for amusement, decoration on novelty items and magazine covers.

Safe Handling of Lasers:

- Only trained, authorized personnel may operate lasers. Authorization is received from the authorized laser user and the Laser Safety Officer.
- NEVER put yourself into any position where your eyes approach the axis of a laser beam (even with eye protection on).
- Keep beam paths below or above standing or sitting eye level. Do not direct them towards other people.
- Do not damage laser protective housings, or malfunctioned the interlocks on these housings.
- Eliminate all reflective material from the vicinity of the beam paths.
- Never use viewing instruments to look directly into a laser beam. If this is necessary, install an appropriate filter into the optical element assembly.
- Keep ambient light levels as high as operations will permit.
- Do not work alone when performing high power laser operations.